Exploring the Impact of Behavioral and Contextual Factors on School Shootings (1966–2023)

Jan 18 , 2026

Fatma Mahfoud

Abstract

School shootings remain a persistent public safety concern in the United States, producing severe human and social costs. This study uses the CHDS School Shooting Safety Compendium (1966–2023) to explore historical trends, describe incident severity distributions, examine associations among behavioral and contextual indicators, and evaluate the feasibility of predictive modeling for injury outcomes. The analysis combines exploratory data analysis, correlation assessment, and supervised learning (bagging and boosting) with techniques to address class imbalance. Results show a long-run increase in reported incidents, highly right-skewed severity distributions, and modest associations among selected indicators. Predictive performance is limited, especially for rare high-severity outcomes, highlighting the difficulty of forecasting extreme events. Findings are framed as exploratory and prevention-oriented, with careful attention to ethical considerations such as privacy, bias, and the risk of stigmatization.

Introduction

School shootings represent a continuing challenge for public safety, education systems, and communities. Beyond immediate loss of life, these events produce long-term trauma and disrupt learning environments. Although many prevention efforts exist, responses are often reactive rather than proactive. This project asks what patterns and associations can be observed in historical incident data, and what they imply for prevention-focused policy and practice. The goal is not to establish causality or to profile individuals, but to provide descriptive and exploratory evidence that can inform earlier intervention, resource allocation, and risk communication.

Data and Dataset Preparation

The analysis uses the CHDS School Shooting Safety Compendium (1966–2023). Variables include incident date and location, injuries and fatalities, number of firearms brought to the scene, and behavioral indicators such as psychiatric medication, paranoia, isolation, notably depressed mood, and childhood trauma. Data preparation included selecting variables with sufficient coverage, converting numeric fields, handling missing values, and creating derived measures such as total casualties.

Class imbalance was addressed using SMOTE during model training, recognizing that rare event prediction remains fundamentally difficult.

Methods

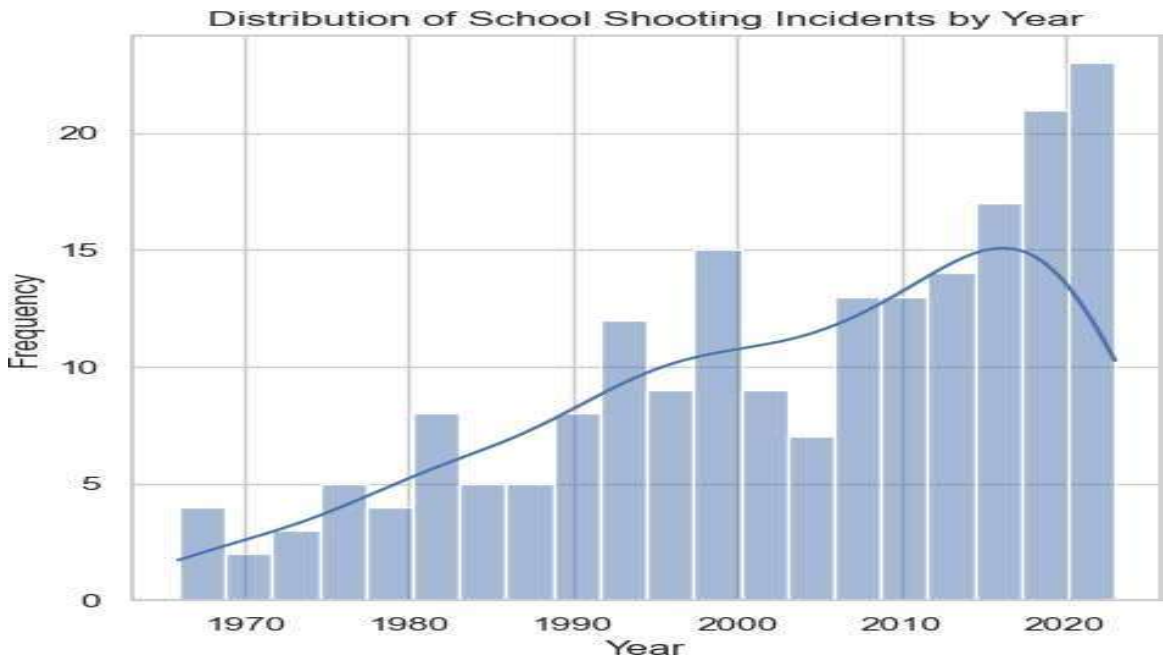
Methods included exploratory data analysis using histograms, boxplots, and correlation heatmaps, followed by supervised learning using bagging (Random Forest) and boosting (Gradient Boosting). Pipelines included imputation, class weighting, train-test splitting, and evaluation using accuracy, macro-averaged precision, recall, F1-score, and confusion matrices. These models were used exploratorily to assess whether selected features contain predictive signal.

Visualizations

To effectively communicate findings, various visualization techniques were employed.

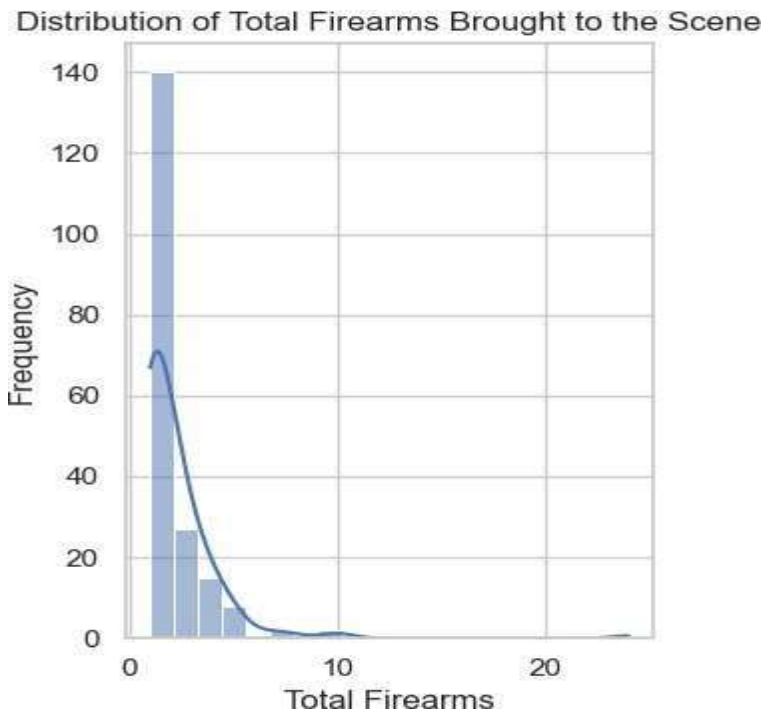
The histogram (below) displays the distribution of school shooting incidents over the years, with the following observations:

1. Overall Increase: There is a noticeable upward trend in the frequency of school shooting incidents from the 1970s to the 2020s. This suggests that such incidents have become more common over the decades.
2. Notable Peaks:
a. The years with the highest frequency appear to cluster towards the 2000s and 2010s, indicating that school shootings may have become a more significant issue during these periods.
b. The highest frequency peaks seem to occur around 2010 and 2018, which could correlate with specific events that gained media attention or changes in societal factors.
3. Recent Decline: There seems to be a slight decline in the frequency of incidents from around 2018 to 2020, which may indicate a temporary reduction in the number of reported cases during those years.
4. KDE Line: The kernel density estimate (KDE) line helps visualize the distribution's shape. The smooth curve shows a consistent increase in incidents over time, reinforcing the idea that school shootings have become more frequent.



The histogram (below) illustrates the distribution of the total number of firearms brought to the scene during school shooting incidents. Here are the key observations:

1. **Skewed Distribution**: The distribution is highly skewed to the right, indicating that most incidents involved a low number of firearms, while a few incidents involved significantly more.
2. **High Frequency at Low Counts**:
    a. The majority of incidents (around 0 to 5 firearms) show a high frequency, suggesting that many shooters brought only a few firearms to the scene.
    b. The frequency drops sharply as the number of firearms increases, indicating that incidents with larger numbers of firearms are much less common.
3. **Few Outliers**: There are a few incidents with a notably higher count of firearms (above 10), but these are rare compared to the overall frequency. This suggests that while multiple firearms can be brought to the scene, most incidents do not involve a large arsenal.
4. **K DE Line**: The kernel density estimate (KDE) line reflects the distribution's shape and corroborates the histogram's observations, showing a peak near the lower end of the firearm count.

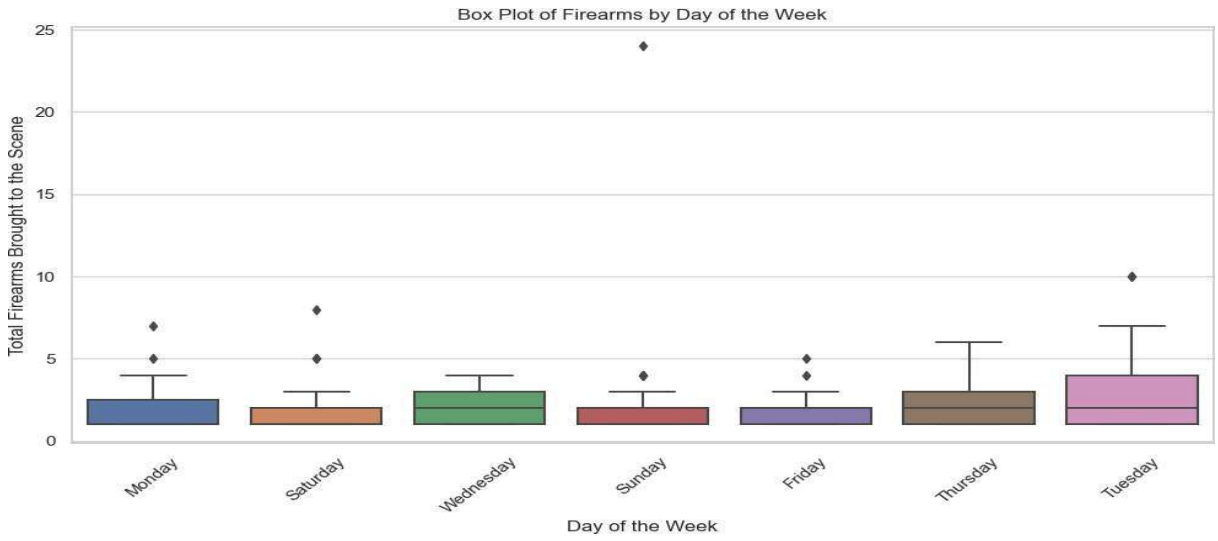


The box plot (below)displays the distribution of the total number of firearms brought to the scene during school shooting incidents, segmented by the day of the week. Here are the key observations:

1. Median and Quartiles:
    a. Each box represents the interquartile range (IQR), which contains the middle 50% of the data. The line inside each box indicates the median number of firearms brought on that day.
    b. The median varies significantly among different days, with some days showing higher median counts than others.
2. Highest Median:
    a. **Monday** shows the highest median, suggesting that incidents occurring on this day tend to involve more firearms compared to other days.
3. Variability Across Days:
    a. **Saturday** and **Thursday** also have relatively high medians, indicating that there are notable incidents on these days as well.

b. **Sunday** and **Tuesday** appear to have lower medians, suggesting fewer firearms are typically involved in incidents occurring on these days.

4. Outliers:
   a. The presence of dots beyond the whiskers indicates outliers—individual incidents where the number of firearms brought to the scene was significantly higher than the IQR. These outliers are more pronounced on certain days, indicating potentially unique incidents that merit further investigation.
5. Spread of Data:
   a. The spread (length of the boxes and whiskers) varies, with some days showing a wider spread in the number of firearms, indicating more variability in incidents.



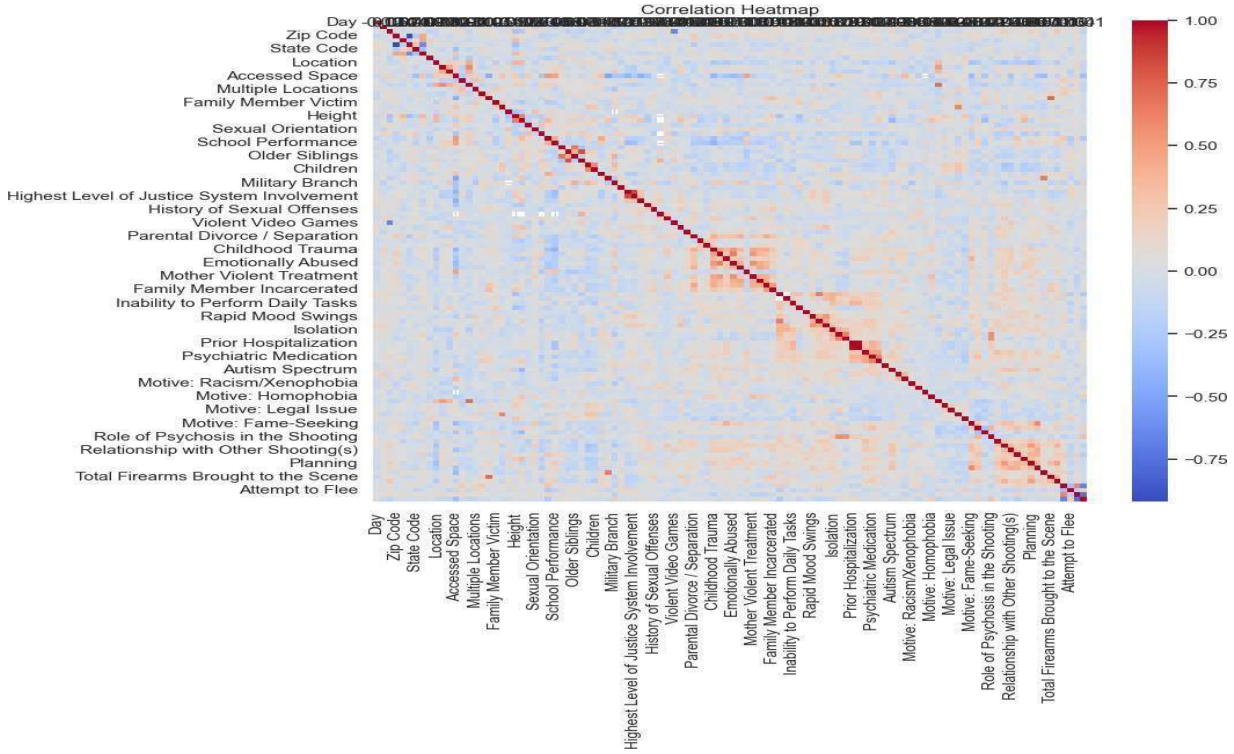
The correlation heatmap (below) shows the relationships between various features in your dataset related to school shootings. Here are some key observations:

1. Color Scale:
   a. The values range from -1 to 1, where values close to 1 indicate a strong positive correlation, values close to -1 indicate a strong negative correlation, and values around 0 suggest little to no correlation.
2. Diagonal Line:
   a. The diagonal line from the top left to the bottom right represents the correlation of each variable with itself, which is always 1.0.
3. Positive Correlations:
   a. Certain features, such as Total Firearms Brought to the Scene and Motive: Psychosis in the Shooting, may show a positive correlation with other variables, suggesting that

incidents involving a higher number of firearms are associated with specific motives or characteristics.

b. Features related to Family (e.g., Family Member Involvement) might also show positive correlations with variables linked to the shooter's background.

4. Negative Correlations:
   a. Some variables may exhibit negative correlations, indicating that as one feature increases, another decreases. For instance, there may be a negative correlation between Inability to Perform Daily Tasks and School Performance, suggesting that poorer performance in school relates to higher difficulties in daily functioning.
5. Clusters of Correlation:
   a. The heatmap may reveal clusters of variables that are closely related, such as those pertaining to mental health, family background, and incidents involving firearms. This could indicate shared underlying factors or trends worth investigating further.
6. Complex Relationships:
   a. The heatmap allows for quick visual identification of complex relationships among multiple variables, which can guide further analysis and research into the factors contributing to school shootings.



Results

Descriptive analysis shows an increase in reported incidents over time, highly right-skewed distributions of firearms and injuries, and differences in medians and variability by day of week. Correlation patterns indicate modest associations among some behavioral and contextual variables. Modeling results show limited predictive performance, particularly for higher injury counts, with confusion matrices revealing frequent misclassification of rare severe cases.

Discussion

Results suggest that school shootings are shaped by multiple interacting factors rather than a single driver. While descriptive patterns are informative, predictive modeling is constrained by class imbalance, data sparsity for extreme events, and the complexity of human behavior. Findings should be interpreted as exploratory and prevention-oriented, not as tools for individual prediction or profiling.

Limitations

This analysis assumes reasonable accuracy and consistency in the CHDS dataset. Limitations include missing and unevenly reported variables, observational data that do not support causal inference, severe class imbalance, and potential reporting bias over time. Results should not be used for individual risk scoring or mental health attribution.

Ethical Considerations

Ethical priorities include privacy protection, avoiding stigmatization of mental health conditions, preventing misuse of results for profiling, and transparent communication of uncertainty and limitations. Findings are framed for systems-level prevention and policy discussion rather than individual-level judgment.

Conclusion

This study provides an exploratory view of long-term trends and severity patterns in school shooting incidents and highlights the difficulty of predicting rare extreme events. Prevention should emphasize multi-layered strategies including mental health support, threat assessment, training, and coordinated community response. Future work should integrate policy, socioeconomic, and geographic variables and use

methods designed for rare event analysis.

Responses to Anticipated Audience Questions

1) **How reliable are behavioral indicators in this dataset?**

Behavioral indicators are unevenly reported and should be treated as imperfect proxies, useful for exploratory patterns but not individual inference.

2) **What evidence supports (or contradicts) links between specific factors and incident severity?**

Correlation analysis shows modest associations, supporting a multi-factor view rather than a single-cause explanation.

3) **How do we prevent stigmatizing mental health conditions when communicating findings?**

Stigmatization is avoided by careful language, avoiding causal claims, and focusing on systems level prevention.

4) **What important variables are missing (e.g., policy context, access to services)?**

Missing variables include policy context, access to services, and community socioeconomic factors.

5) **Are trends increasing, or is reporting improving over time?**

Trends likely reflect both real changes and improved reporting over time.

6) **How should schools act on findings without overreacting or misapplying them?**

Schools should use findings to support layered prevention strategies, not profiling.

7) **How generalizable are results beyond the dataset scope and definitions?**

Results are limited to this dataset and may not generalize to other contexts.

8) **What safeguards prevent misuse for individual profiling?**

Safeguards include clear framing, transparency about limits, and avoidance of

individual-level predictions.

9) **How should uncertainty and limitations be communicated to non-technical audiences?**

Uncertainty should be communicated in plain language with visual summaries and explicit limitations.

10) **Which prevention approaches are supported by the strongest evidence?**

Strongest evidence supports multi-component prevention strategies rather than single interventions.

References

Center for Homeland Defense and Security. (2023). School Shooting Safety Compendium.

Rocque, M. (2012). Exploring school rampage shootings: Research, theory, and policy. The Social Science Journal, 49(3), 304–313.

Swearer, S. M., Espelage, D. L., Vaillancourt, T., & Hymel, S. (2010). What can be done about school bullying? Linking research to educational practice. Educational Researcher, 39(1), 38–47.

Wintemute, G. J. (2015). The epidemiology of firearm violence in the twenty-first century United States. Annual Review of Public Health, 36, 5–19.

https://github.com/Fatma-Amel/Exploring-the-Impact-of-Behavioral-and-Contextual-Factors-on-School-Shootings-1966-2023-/tree/main